

```
1 CREATE DATABASE Customer_Demographics;
2
3 USE Customer_Demographics;
4
5 SELECT TOP 10 *
6 FROM customer_segementation_Cleaned_data;
7
8 --What is the totla number of customers?
9 select count(*) as Total_Customer
10 from customer_segementation_Cleaned_data
11
12
13
14 --What is the Average age of customer?
15 select avg(age) as Customer_Age
16 from customer_segementation_Cleaned_data
17
18
19
20 --What is the Gender Distribution?
21 select gender,
22 count(*) as total_customer
23 from customer_segementation_Cleaned_data
24 group by gender
25 order by total_customer
26
27
28
29 --Which age group has the highest average income?
30 select top 1
31 case
32 when age between 18 and 25 then '18-25'
33 when age between 26 and 35 then '26-35'
34 when age between 36 and 45 then '36-45'
35 when age between 46 and 55 then '46-55'
36 else '56+'
37 end as age_group,
38
39 ROUND(AVG(income), 2) AS average_income
40
41
42
43 from customer_segementation_Cleaned_data
44
45 group by
46 case
47 when age between 18 and 25 then '18-25'
48 when age between 26 and 35 then '26-35'
49 when age between 36 and 45 then '36-45'
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50 when age between 46 and 55 then '46-55'
51 else '56+'
52 end
53
54 order by Average_Income desc
55
56
57 --Which preferred category has the most customers?
58 select preferred_category,
59 count(*) as total_customer
60 from customer_segementation_Cleaned_data
61 group by preferred_category
62 order by total_customer desc
63
64
65 --What is the Average Income by gender?
66 select gender,AVG(income) as average_income
67 from customer_segementation_Cleaned_data
68 group by gender
69 order by average_income desc
70
71
72 --Which customer segment spends the most?
73 select preferred_category,
74 AVG(spending_score) as avg_spending_score
75 from customer_segementation_Cleaned_data
76 group by preferred_category
77 order by avg_spending_score desc
78
79
80 --How does purches frequency vary by age group?
81 select
82 case
83 when age between 18 and 25 then '18-25'
84 when age between 26 and 35 then '26-35'
85 when age between 36 and 45 then '36-45'
86 when age between 46 and 55 then '46-55'
87 else '56+'
88 end as age_group,
89
90 ROUND(AVG(purchase_frequency), 2) as AVG_purchase_frequency
91
92 from customer_segementation_Cleaned_data
93
94 group by
95 case
96 when age between 18 and 25 then '18-25'
97 when age between 26 and 35 then '26-35'
98 when age between 36 and 45 then '36-45'
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99  when age between 46 and 55 then '46-55'
100 else '56+'
101 end
102
103 order by AVG_purchase_frequency desc
104
105
106 --Which product/category is preferred by different demographics?
107 select
108 case
109 when age between 18 and 25 then '18-25'
110 when age between 26 and 35 then '26-35'
111 when age between 36 and 45 then '36-45'
112 when age between 46 and 55 then '46-55'
113 else '56+'
114 end as age_group,
115
116 preferred_category,
117
118 count(*) as total_customer
119
120 from customer_segementation_Cleaned_data
121
122 group by
123 case
124 when age between 18 and 25 then '18-25'
125 when age between 26 and 35 then '26-35'
126 when age between 36 and 45 then '36-45'
127 when age between 46 and 55 then '46-55'
128 else '56+'
129 end,
130 preferred_category
131
132 order by total_customer desc
133
134
135 --Which demographic group represents the most valuable customers
136 select top 1
137 case
138 when age between 18 and 25 then '18-25'
139 when age between 26 and 35 then '26-35'
140 when age between 36 and 45 then '36-45'
141 when age between 46 and 55 then '46-55'
142 else '56+'
143 end as age_group,
144
145 AVG(spending_score) as avg_spending_score,
146
147 ROUND(
```

```
148         AVG(last_purchase_amount * purchase_frequency),
149         2
150     ) AS estimated_customer_value
151
152 from customer_segementation_Cleaned_data
153
154 group by
155 case
156 when age between 18 and 25 then '18-25'
157 when age between 26 and 35 then '26-35'
158 when age between 36 and 45 then '36-45'
159 when age between 46 and 55 then '46-55'
160 else '56+'
161 end
162
163 order by estimated_customer_value desc
164
165
166
167
```